Fiscal Year:	FY 2007	**Task Last Updated:**	FY 11/13/2007
PI Name:	Putcha, Lakshmi Ph.D.		
Project Title:	Assessment of Pharmaceutical Stability in Space Using Ground-based Simulation of Spacecraft Environmental Factors		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:	Deceased as of September 2015.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	05/15/2004	**End Date:**	06/01/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	3
No. of PhD Candidates:	0	**No. of Master' Degrees:**	2
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Du, Brian (Wyle Laboratories)		
Grant/Contract No.:	not available		
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Performance Goal Text:			
	Therapeutic efficacy and safety of pharmaceuticals flown on the space shuttle and International Space Station (ISS) remain critical issues for successful NASA medical operations. Preliminary studies indicated significant differences in potency of some pharmaceutical preparations flown on space flights compared to ground-based controls. This project addresses Question number 11 of Clinical Capabilities in the Critical Path Roadmap. We hypothesize that space flight environment, especially radiation, affects stability of pharmaceuticals contained in the space shuttle and ISS medical kits. The objectives are to 1) determine chemical and physical stability parameters of select pharmaceuticals contained in the shuttle and ISS medical kits along with matching ground-based controls; 2) characterize degradation profiles of the unstable formulations after exposure to particulate radiation using synchrotron light and radiation sources; and 3) develop mathematical models that can predict shelf life of pharmaceuticals during space flight.		

Task Description:	Select dosage forms from the medical kits returned to Earth after a space flight will be analyzed and compared to matching formulations stored under controlled conditions on Earth. Disintegration, dissolution rates and concentrations of active drug and degradation products of these dosage forms will be determined. Effects of high-energy radioactive particles on the stability of these pharmaceuticals will be examined by exposing the formulations to synchrotron light and radiation sources at Brookhaven National Laboratories. Predictive degradation models will be constructed from these data to estimate the degree and magnitude of loss of pharmaceutical activity during space flights. In addition to providing information on pharmaceutical stability in space, the proposed research will also identify if there is a need to develop special pharmaceutical preparations that can ensure stability and resultant therapeutic efficacy of pharmaceuticals in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research expands and adds to current understanding of medication and dosage form stability. The primary customer inside NASA is the astronaut flight crew and Flight Medicine by improving and optimizing medical kit configuration. Understanding medication stability in spaceflight benefits the crew by ensuring treatment success. The primary application of this research outside NASA, primarily by optimization of medicine storage or optimization of medical kit configuration in special environments such as military, maritime, and remote and isolated research outposts.
Task Progress:	Stability of drugs on the ground is known to be dependent on a number of environmental factors including but not limited to exposure to heat, light, and humidity, as well as other storage conditions. Some medications used in spaceflight have been used in doses far higher than those used on the ground or have produced unusual side effects, both of which can be explained by alterations in drug stability. Drugs are stored in manners that aer not conventional on the ground, and outside the normal stability rated storage conditions manufacturers report to the FDA. In addition, objects in space are exposed to background cosmic radiation and to radiation from Solar Particle Events (SEPs) both of which contain ionizing radiation or heavy ions. The radiation dose inside spacecraft, however, is higher than without due to production of daughter ions and this increase can cause a reduction in the stability of drugs. In this experiment, we examine the total content of active drug in each dosage form as well as other physical properties which may affect drug efficacy, such as dissolution rate, which may be affected by radiation or other environmental factors. This experiment is divided into two separate but related arms, a flight payload experiment and a ground experiment conducted at the National Space Radiation Laboratory, both of which were initiated in 2006. In the flight arm of this experiment a payload of several identically packed medication kits containing 35 dosage forms comprised of 29 prescription or over the counter medications was launched onboard STS-121 on July 4, 2006, and transported to the International Space Station. Over a planned period of 18 months, individual packed medication kits will be returned from the ISS and medications analyzed for total drug content and dissolution rate. On the ground, drugs are regulated by the FDA to deliver within 5% of the labelled claim of the dose for a predetermined time period which is not to exceed two years. To date, the kits returned have revealed stability decrements for several drugs. Several medications were found to have less than 95% of their label claim when flown on STS-121; they are clavulanate (in Augmentin), levothyroxin, and risedronate. Medications which were flown onboard STS-121 were also compared to identically packed kits which were left in controlled storage on the ground at Kennedy Space Center. Medications which showed a greater than 5% difference in active drug content compared to KSC controls were furosemide and cilastatin (in Primaxin). Similarly, in our ground based study we found decrements in overall drug content when exposed to iron (Fe) and proton (p+) beams ranging from 0.1 to 50 Gray, doses which although higher than terminal doses to humans are suitable for accelerated stability testing as defined by the FDA. Some drugs failed to meet the expected 95% of dose in this scenario were acetaminophen, cavulanate (in Augmentin), ciprofloxacin (in ophthalmic solution), mupirocin, phenazopyridine, and promethazine. Compared to controls, two drugs had a greater than 5% reduction in content, clavulanate (in Augmentin) and phenazopyridine. In addition to drug content delivered, other factors such as dissolution rate can be affected by environmental factors. Preliminary results from this single exposure to spaceflight on STS-121 indicates dissolution rates for several drugs were different from the rates of ground controls; the drugs with reduced release rates after exposure to spaceflight were azithromycin, ciprofloxacin, levothyroxin, folate (in a multivitamin), and niacin (in a multivitamin). Several drugs exposed to radiation exposures at NSRL also had reduced dissolution rates; they are amoxicillin & clavulanate, trimethoprim & sulfmethoxazole, ciprofloxacin tablets, ibuprofen. Three drugs had reduced release rates when exposed to Fe beams but not p+ beams; they are levothyroxin, mupirocin, and riboflavin. Analysis of drug content and dissolution rates in future payload returns, the next of which is expected in June 2007, and in further exposures at NSRL, is required to ascertain the true effect of spaceflight on drugs.
Bibliography Type:	Description: (Last Updated: 06/23/2025)
Abstracts for Journals and Proceedings	Daniels V, Putcha L. "Pharmacovigilance in Space: Stability Payload Compliance Procedures." NSBRI/NASA Human Research Program 2007 Annual Meeting, Seabrook, TX, February 2007. NSBRI/NASA Human Research Program 2007 Annual Meeting, February 2007. , Feb-2007
Abstracts for Journals and Proceedings	Daniels V, Putcha L, McCluskey R. "Pharmacovigilance in Space." American College of Clinical Pharmacy 2006 Annual Meeting, October, 2006. ACCP Meeting Proceedings, October 2006. , Oct-2006